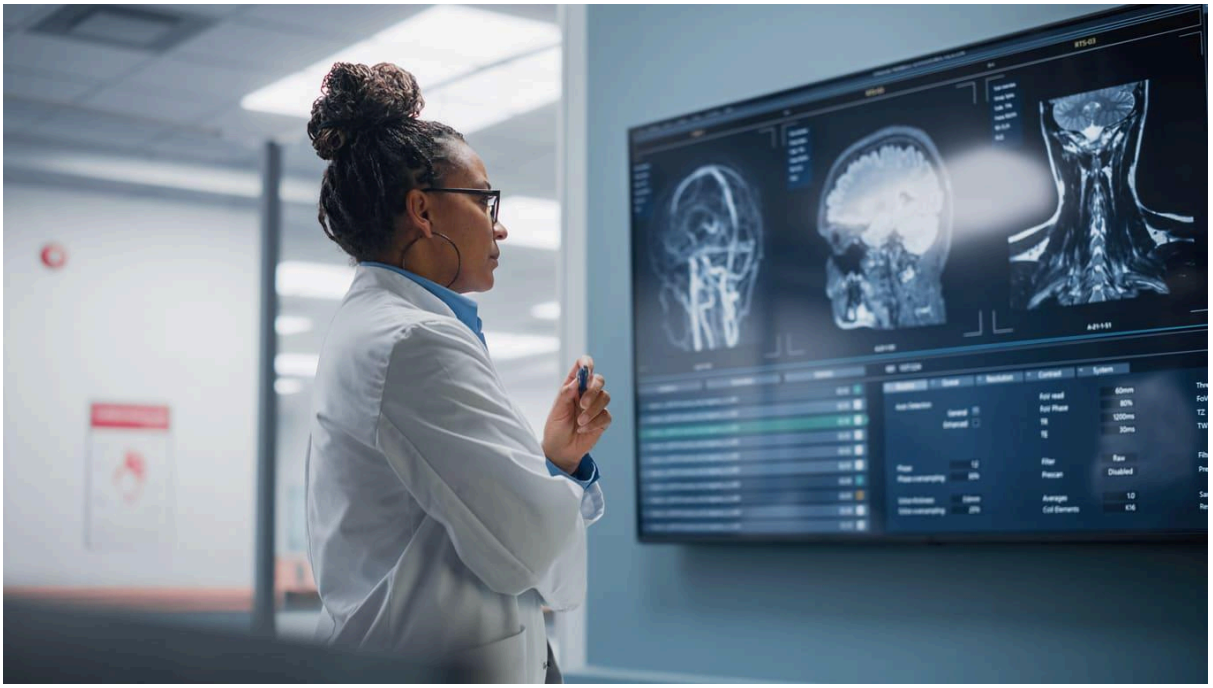
Phase 1 – Computer Vision

Project Planning Report

AI-Based Brain Tumour Detection



Group Members:

Oluwafemi Gabriel James, Marrium Atique, Kevin Kovacik

Table Of Contents:

Project Planning Report **1**

AI-Based Brain Tumour Detection **1**

1. Introduction **2**

2. Business Understanding **3**

2.1 Problem Definition 3

2.2 Business Questions 4

2.3 Business Goals 4

3. Data Understanding **4**

3.1 Dataset Overview 4

3.2 Data Quality and Observations 5

3.3 Data Limitations and Mitigation Strategies 6

4. Data Preparation **7**

4.1 Preprocessing Steps 7

4.2 Data Enrichment 8

5. Modeling and Testing **8**

5.1 Model Architecture Options 8

5.2 Model Testing and Validation 9

5.3 Expected Output 9

6. Evaluation and Deployment **10**

6.1 Evaluation 10

6.2 Deployment Plan 10

6.3 Maintenance and Future Improvements 10

7. Conclusion **10**

Reference for the Project Planning Report **11**

1. Introduction

The rapid evolution of **AI and Computer Vision** has opened new frontiers in healthcare diagnostics, enabling earlier detection, reduced diagnostic errors, and faster medical decision-making. This project applies **deep learning and image recognition** techniques to **brain MRI scans** for the automated detection of brain tumors.

This project applies the **CRISP-DM (Cross-Industry Standard Process for Data Mining) framework to design, develop, and evaluate a computer vision-based brain tumor classification system**. The model leverages **Convolutional Neural Networks (CNNs)** to classify MRI images into two categories:

- **YES** → Tumour present
- **NO** → No tumour detected

The model integrates **YOLOv8 (via Roboflow/GVision)**, a state-of-the-art object detection and image classification framework, with **transfer learning**, **data augmentation**, and **cross-validation** to create a robust diagnostic pipeline.

The project's methodology includes:

- **Data ingestion and exploration** of MRI images
- **Data augmentation** to increase dataset diversity
- **Model development** using a pretrained architecture (e.g., YOLOV8)
- **Evaluation** with accuracy, precision, recall, F1-score, and confusion matrix
- **Deployment simulation** through a model inference pipeline for unseen test images

Technologies used: Python, PyTorch, Roboflow, YOLOv8, TensorFlow/Keras, OpenCV

2. Business Understanding

2.1 Problem Definition

Brain tumor diagnosis using MRI scans is labor-intensive, time-sensitive, and highly dependent on radiologist expertise. Misinterpretation or late detection can lead to delayed treatment and adverse outcomes.

This project addresses the challenge by developing an **AI-assisted diagnostic system** that can analyze MRI images and **automatically detect the presence of tumors**, serving as a clinical decision-support tool.

2.2 Business Questions

1. Can deep learning models reliably detect brain tumors in MRI images with limited training data?
2. How does transfer learning improve accuracy and reduce data dependency?
3. What preprocessing and augmentation techniques enhance model generalization?
4. Can model interpretability (via Grad-CAM) increase clinical trust and adoption?
5. How can this system be deployed efficiently for real-world diagnostic support?

2.3 Business Goals

1. **Accuracy Goal:** Achieve ≥90% classification accuracy using transfer learning.
2. **Scalability Goal:** Develop a modular model that can be retrained with new data.
3. **Interpretability Goal:** Implement visual explanations for tumor detection zones.
4. **Efficiency Goal:** Reduce diagnostic time while maintaining clinical reliability.
5. **Deployment Goal:** Deliver a functional prototype accessible via a web interface.

3. Data Understanding

3.1 Dataset Overview

- **Primary Source:** Kaggle – Brain Tumour Dataset (Yes/No)
- Roboflow integration for pretrained model weights.

Structure:

```
brain_tumor_df/
    └── BarinTumorDataSet/
        └── brain_tumor_dataset/
            ├── yes/
            └── no/
```

- **Data Type:** JPG, JPEG, PNG images
- **Classes:** Binary classification
    - yes - Brain tumor detected
    - no - No brain tumor
- **Total Images:** ~ 253 files
    - yes - 155 files
    - no - 98 files
- **Image Dimensions:** Variable (resized to 224x224 for model input)
- **Data Type:** MRI brain scans, RGB, variable contrast.

3.2 Data Quality and Observations

- **Image Variation:** Image clarity varies; some images show strong contrast, while others have noise or blur.
- **File Naming Irregularities:** Filenames vary in format (e.g., *Y1.jpg*, *no 10.jpg*, *N17.JPG*), indicating manual or multi-source data collection.
- **Image Quality:** Background intensity differs across MRI sources.
- **Class Distribution:** The dataset is **slightly imbalanced** (≈60% tumour, 40% no tumour), but both categories have sufficient samples for binary classification.
- **Consistency:** The data structure is clean (two folders representing the classes), making it straightforward for supervised learning pipelines.
- No evident missing files, but duplicate checks are required.

3.3 Data Limitations and Mitigation Strategies

While the dataset is organized and suitable for binary classification, several constraints exist:

s/n	Limitations	Reasons	Mitigation Strategies
1	**Small Dataset Size**	With ~253 images, training a CNN from scratch risks overfitting	• Use **YOLOv8 pretrained weights (via Roboflow/GVision)** as a feature extractor. • Apply **transfer learning** to fine-tune the top layers on our MRI dataset. • This approach leverages learned image features (edges, gradients, textures) from large-scale datasets.
2	**Limited MRI Diversity**	Images likely originate from limited scanner types or sources.	• Employ **data augmentation** (rotation, brightness, zoom, flips). • Leverage YOLOv8's **broad feature generalization** for unseen data robustness.
3	**Potential Duplicates and Naming Issues**	Inconsistent filenames may introduce label confusion.	• Run a **hash-based duplicate check** and enforce consistent naming (e.g., yes_001.jpg).
4	**Model Depth vs. Data Size**	Large CNNs on small datasets may memorize patterns.	• Use **dropout**, **early stopping**, and **frozen YOLOv8 layers** during fine-tuning.

5	**Domain Bias**	MRI scans may not generalize across demographics or scanners.	• Validate on unseen datasets or external open MRI repositories (e.g., TCIA). • Retrain periodically with new hospital data.
6	**Integration and Validation with YOLOv8**	Fine-tune pretrained YOLOv8 layers to adapt feature maps for MRI classification.	Our dataset will serve for: • Fine-tuning the classification head. • Validation/testing. • Comparative benchmarking vs. baseline CNN.

4. Data Preparation

4.1 Preprocessing Steps

s/n	Step	Description
1	**Data Loading**	Import images and assign binary labels ("yes" = 1, "no" = 0).
2	**Resizing**	All images resized to 224×224×3 for CNN input compatibility.
3	**Normalization**	Pixel scaling to [0, 1] to stabilize and accelerate training.

4	**Augmentation**	Random rotation (±15°), flips, brightness/contrast adjustments, zoom, and Gaussian noise injection.
5	**Data Split**	Train (70%), Validation (20%), Test (10%) with stratified sampling.
6	**Balancing**	Maintain near-equal representation across both classes.

4.2 Data Enrichment

To improve feature diversity, additional MRI samples from public repositories such as **TCIA** or **Figshare** may be integrated into the YOLOv8 fine-tuning stage.

5. Modeling and Testing

5.1 Model Architecture Options

s/n	Approach	Description	Strength	Weakness
1	**Baseline CNN**	Custom convolutional model with pooling and dense layers	Simplicity, interpretability	Lower accuracy, prone to overfitting
2	**Transfer Learning (Preferred)**	YOLOv8 pretrained on Roboflow medical images, fine-tuned on the MRI dataset	High generalization, reduced training time	Requires GPU, limited interpretability

Configuration:

- **Pretrained Model:** YOLOv8 (Roboflow/GVision)
- **Frozen Layers:** Backbone feature extractor
- **Trainable Layers:** Classification head (binary)
- **Optimizer:** Adam (lr = 0.0001)
- **Loss:** Binary Crossentropy
- **Metrics:** Accuracy, Precision, Recall, F1-score, ROC-AUC

5.2 Model Testing and Validation

Evaluation Techniques:

- **Confusion Matrix:** Analyze TP/FP/FN rates
- **ROC-AUC Curve:** Assess classification threshold performance
- **Cross-validation:** K-fold validation for generalization
- **Grad-CAM Visualization:** Highlight tumour regions influencing decisions

Performance Goal:

- Accuracy ≥ 90%
- Recall ≥ 85% (priority on sensitivity for tumour detection)

5.3 Expected Output

s/n	Input	Output	Confidence
1	MRI image (.jpg)	"Tumour Detected" / "No Tumour Detected"	0.00–1.00 (probability score)

6. Evaluation and Deployment

6.1 Evaluation

- **Quantitative:** Accuracy, precision, recall, F1, ROC-AUC
- **Qualitative:** Grad-CAM visual heatmaps for interpretability
- **Error Analysis:** Misclassified images reviewed for brightness, blur, or feature overlap issues

6.2 Deployment Plan

- Save the trained model as .h5 or ONNX for portability
- Integrate via a **Streamlit web app,** where users can upload MRI scans
- Backend performs inference and displays:
    - Prediction result
    - Confidence score
    - Grad-CAM tumour region overlay
- Future integration: Cloud hosting on **AWS S3 + EC2** or **Azure ML Studio**

6.3 Maintenance and Future Improvements

- Periodic retraining with new MRI data
- Introduce **multi-class segmentation** (tumour type classification)
- Implement **explainable AI dashboards** for clinical review
- Continuous model monitoring for drift detection

7. Conclusion

This project demonstrates an **end-to-end AI-driven healthcare pipeline** built on the CRISP-DM framework for brain tumour detection. By combining **YOLOv8 pretrained models**, **transfer learning**, and **robust preprocessing**, it overcomes small dataset limitations while maintaining clinical reliability and interpretability.

The hybrid approach, **feature extraction from YOLOv8** and **fine-tuning with local MRI data**, showcases how modern computer vision techniques can be responsibly deployed in healthcare diagnostics to enhance **speed**, **accuracy**, and **decision support** for medical professionals.

Reference for the Project Planning Report

James, O. G., Atique, M., & Kovacik, K. ([2025]) **Phase 1 – Computer Vision: Project Planning Report – AI-Based Brain Tumour Detection** [Unpublished Academic Report].

Key Components

- **CRISP-DM (Cross-Industry Standard Process for Data Mining)** R. Wirth and J. Hipp, "CRISP-DM: Towards a Standard Process Model for Data Mining," in *Proc. 4th Int. Conf. Pract. Appl. Knowl. Discovery Data Mining (PAKDD '00)*, 2000.
- **YOLOv8 (You Only Look Once v8)** Ultralytics. (**[2025]**) **YOLOv8.** [Computer Software]. [Online]. Available: https://github.com/ultralytics/ultralytics
- **Roboflow (For dataset management and transfer learning)** B. Dwyer, J. Nelson, T. Hansen, et al. (**[2025]**) **Roboflow (Version 1.0)** [Software]. Available: https://roboflow.com
- **Kaggle Brain Tumour Dataset (Yes/No)** Ihtisham Khattak. (**[2025]**) **Brain Tumor Dataset.** [Online]. Available: https://www.kaggle.com/datasets/ihtishamkhattak99/brain-tumor-dataset-yes-no